YOJO CYBER ACADEMY · BACKEND

Open items and decisions

Everything that is still open in the backend, grouped by what it is waiting for: 65 items from docs/OPEN-ITEMS.md.

Status	Items	What it means
Blocked on a provider or account	7	The code is ready behind an interface; it needs a service, key or account that Yojo has not supplied yet.
Needs a Yojo decision	12	Something is built with a provisional default or left unchanged on purpose. Confirm the default or tell us the change.
Can be built now	4	No outside dependency; these can be scheduled as development work.
Before production	4	Needed before real students use the platform; not needed for local development.
Decided, done or recorded for information	38	Decided by the owner, built, or an assumption documented in code. No action unless Yojo wants a change. The owner's 24 Oct decisions come first.

Prepared 26 October 2026. Item IDs (OI-nn) match docs/OPEN-ITEMS.md. Green "Update" lines are status changes since the item was first written. Updated with the owner's answers of 24 Oct.

Blocked on a provider or account 7

The code is ready behind an interface; it needs a service, key or account that Yojo has not supplied yet.

ID	Item	Details
OI-01	**Transactional email provider (OD-11)** Platform	`MAIL_DRIVER=dev` writes emails to `.dev-mail/`; `disabled` drops them. Production refuses `dev`, so OTP, invites and resets cannot be delivered in production yet. Provider choice + API key; add an adapter implementing `MailPort` (`src/identity/mail/mail.port.ts`).
OI-02	**Google sign-in client ID** Platform	`POST /auth/google` answers 503 `GOOGLE_NOT_CONFIGURED`. Verification code (JWKS, iss, aud, exp, email_verified) is written but not exercised against Google. Google OAuth client ID in `GOOGLE_CLIENT_ID`; then add an e2e test with a real token in staging.
OI-03	**Cloud, KMS, secrets manager (OD-01)** Platform	`FIELD_ENCRYPTION_KEY` comes from env. Key IDs are stored so rotation is possible; no re-encryption job yet. Cloud account; KMS data key injection; rotation runbook (AC-ENV-04).
OI-21	**Paid plans and prices.** Auth / CTF / Admin	Only the `free` plan is seeded. Paid access = live entitlement `plan:paid` (granted manually by a Platform Admin with reason + fresh MFA, or later by payments), or a plan whose `grants` contains it. Prices, plan list and the payment webhook need credentials and a decision.
OI-36	**Notifications.** Public website	Acknowledgement to the visitor, alert to `ENQUIRY_NOTIFY_EMAIL`, notice to the assignee: queued in the outbox and sent by the worker through the mail port. With `MAIL_DRIVER=dev` they are files in `.dev-mail`, not real delivery (OI-01). No SLA reminder job yet.
OI-45	**Storage and PDF. Lesson files, certificate PDFs, resume PDFs.** Academy / LMS	Responses say `STORAGE_NOT_CONFIGURED` / `PDF_NOT_CONFIGURED`. Needs an object store (private bucket, signed URLs, malware scanning) and a PDF renderer.
OI-46	**Orbi assistant.** Academy / LMS	No AI provider yet: `POST /student/assistant/messages` answers 503 `ASSISTANT_NOT_CONFIGURED` and Orbi says so plainly. **Updated 26 Oct:** Orbi's built-in help now follows the page (dashboard, labs, leaderboard, Intel, career path, attendance, assessments, placement, learning, Training Grid, courses): its greeting, quick buttons and answers change per section and use the student's own data from the existing APIs (labs cleared and next lab, XP and streak, attendance %, assessments due, readiness, league zone). The unapproved placement-assurance line (OI-31) and the "leagues of about thirty" line were removed. Still needed for free chat: a provider and a data policy (no prompt logging of personal data).

Needs a Yojo decision 12

Something is built with a provisional default or left unchanged on purpose. Confirm the default or tell us the change.

ID	Item	Details
OI-65	**Program pages and internships: draft content to confirm (owner, 30 Oct).** Academy / LMS	Built: each "Our programs" card on the Internships page opens its own page with its internships, details and an application form that reaches Admin → Enquiries tagged "program · internship" (the chosen duration and mode are in the message). **Owner's details (30 Oct):** cyber and AI internships by category; durations 1, 3 or 6 months; online and offline, offline preferred. **Drafted, for Yojo to confirm or correct:** the internships per program (Expert: SOC Analyst, VAPT, GRC Analyst, Digital Forensics, Threat Intelligence, AI Security; Diploma: Cyber Security Foundations, Junior SOC, Web Security, GRC Documentation, AI for Cyber Security; GRC Mentorship: ISO 27001 Implementation, Risk Assessment, Audit Support, Business Continuity, Privacy Compliance, AI Governance), descriptions, which durations each offers (advanced roles VAPT, Forensics and AI Security: 3 or 6 months), schedule wording, eligibility, certificate wording and selection steps. Not shown until Yojo provides them: fees, stipends, seat numbers and batch dates. Certification cards use themed badges, not official logos (trademarks). "Where our learners get placed" lists industries, not companies; real company logos would need confirmed placements and permission (OI-31).
OI-41	**Hint cost base. The owner's words support two readings: a share of the lab's reward, or a share of the student's total XP.** Academy / LMS	Built as a setting, `gamification.hintBase`, default `lab_reward` (the reward paid at completion is cut by 5/10/15 %, whole numbers). `total_xp` charges the % of total XP at the moment the hint opens. Confirm which one.
OI-52	**Career-path quiz wording (owner, 24 Oct).** Academy / LMS	15 scenarios written as a first draft in `src/academy/career-quiz.ts` (version `career-path-2026-10-v1`), 5 themed per track, one answer per track in every question. Yojo should review the wording before real students take it; a change of wording or mapping needs a new version id.
OI-08	**Legal texts and versions (Terms, Privacy, AUP, parental, Code of Conduct, Student Data Handling).** Platform	Version labels are placeholders `2026-10-draft` (`src/identity/constants.ts`). Stored with each consent so they can be re-asked when the texts change.
OI-34	**Legal texts. Footer Terms, Privacy and Refund open the placement-terms text. Consent versions for forms and subscriptions are placeholders `2026-10-draft`.** Public website	Waiting for the legal texts (see OI-08). Applicable Indian privacy law (DPDP Act 2023) needs review by Yojo's legal owner; a consent box alone is not compliance.

ID	Item	Details
OI-09	**How Yojo-origin student accounts are created (bulk SIS import?).** Platform	Platform Admin creates them one at a time with `POST /admin/students` (origin `yojo`, optional student ID). The YES gate accepts a Yojo email or student ID of any existing student; unknown IDs get the same 202 with no email and end in the generic `YOJO_ACCOUNT_NOT_FOUND`. Bulk import is not built.
OI-10	**Required sign-up fields.** Platform	Now required: name and an institution (`institutionId` from the list or free-text `institutionName`); city optional. Domain rule enforced for listed institutions. Other profile fields are collected in onboarding (`/me/missing-fields`).
OI-13	**Session lifetimes (OD-24) and the rule "a change of both IP and user agent ends a staff session" [PROVISIONAL].** Platform	Lifetimes are env-configurable with the spec defaults. The IP + user-agent rule is not implemented yet.
OI-19	**Rate-limit values** Platform	Implemented exactly as the §22.3 table [PROVISIONAL]; NGINX global limits are outside this code.
OI-35	**Retention of enquiries and subscriptions.** Public website	No automatic deletion yet. History and notes are append-only for updates; deletes stay possible so an erasure job can be added once Yojo sets the retention period.
OI-39	**Mentorship form has no email field, so no acknowledgement email can be sent for it (the page used to promise one).** Public website	The success text now gives the reference and says the team replies within one working day. Add an email field if confirmations by email are wanted.
OI-44	**Practice missions in live mode (owner decision, 24 Oct).** Academy / LMS	The in-browser engines M1–M9 stay playable in live mode, labelled "Demo — XP not saved". They talk only to `mission-api`; nothing reaches the XP ledger, streak, daily patches or league. Switch them to the server catalog when OI-43 is done.

Can be built now 4

No outside dependency; these can be scheduled as development work.

ID	Item	Details
OI-58	**Four admin screens stay on demo data (owner, 24 Oct): Dashboard, Student Management, System Settings, CTF Registrations. Wire them when the remaining services exist.** Academy / LMS	Backend status: Settings complete (versioned, audited sections). Student Management mostly there (list, detail, add student, enrol, plan entitlements, suspend, sign-out, XP corrections); missing admin edit of student details (institution, city, learning mode with reason, internal note) and enrolment remove/transfer. CTF Registrations: list, detail and status exist; missing edit of registration details and a per-user login history. Dashboard: counts exist; the paid-access approval queue needs the payments service.
OI-47	**Broadcast second admin.** Academy / LMS	**Buildable now: dual control for large broadcasts, scheduling, recall and templates (WhatsApp needs a provider).** Broadcasts above `policy.broadcastSecondAdminThreshold` (200 students) are refused with `NEEDS_SECOND_ADMIN`; the dual-control flow is not extended to broadcasts yet. Scheduling, recall, wish templates and WhatsApp are not built.
OI-55	**Analytics history and telemetry.** Academy / LMS	**Buildable now: stored monthly snapshots for trends. Page-usage telemetry needs an event-collection service.** Analytics are computed live from records with definitions; there are no stored snapshots, so no month-over-month trends. Page-usage analytics (sessions, load times, friction) need a server event-collection service; the Insight Engine shows rule-based insights only, no AI feed.
OI-38	**Courses on the public site still come from the demo `YojoCourses` data in `trainer-data.js`.** Public website	**Update: the LMS course catalog now exists, so the public course list can be switched to it.** Course catalog API belongs to the LMS milestone (courses, modules, lessons). The academy "Enroll Now" request goes to the server inbox.

Before production 4

Needed before real students use the platform; not needed for local development.

ID	Item	Details
OI-63	**Sign-in demo (prototype experience), LOCAL ONLY: remove before production (owner, 29 Oct).** Academy / LMS	The login is the prototype's popup again (Student/Employee, Yes/No, New Yojo learner / Institution student, Google demo account picker, staff "Send OTP" then authenticator) and the portals open as full pages; it runs on the real local server. Demo people: Riya and Neha (existing Yojo students), Arjun (new), Vikram (trainer), Priya (Platform Admin), made by `scripts/dev-demo-users.ts`. The "Demo accounts & inbox" panel shows the email codes for *.demo addresses and the demo authenticator codes. Local-only routes: `GET /dev/demo` and the demo staff email-code sign-in `POST /dev/demo/staff/otp` + `/verify` (only Vikram and Priya; real staff keep password + authenticator). All answer 404 unless APP_ENV is local or development (tested). Before going live: delete the demo accounts, `src/dev/demo-directory.ts` and the demo code in `login-live.js`. **Added 30 Oct (demo conveniences, frontend only, remove before production):** (1) sign-in is just the email for demo accounts: the login page reads the one-time code from the local demo inbox and generates the demo staff's authenticator code itself, so no OTP or MFA screen appears (the server still checks both; real accounts see the normal screens); (2) the 30-minute inactivity sign-out timers were removed from the admin, trainer and login pages, and the portals send a light keep-alive request every 4 minutes while open (the server's idle limit is unchanged).
OI-04	**Repository and CI (AC-SEC-01, AC-ENV-03, AC-OPS-04)** Platform	`ci/backend-ci.yml` (copy into `.github/workflows/` of the repository) written but never run. The project folder is not a git repository. Zero-downtime migration test needs a pipeline and a rolling deploy. Create the repository, enable CI, add image signing and the staging pipeline.
OI-05	**Infrastructure as code (AC-ENV-01)** Platform	Only `docker-compose.yml` and a `Dockerfile` for local use. Cloud decision (OD-01), then Terraform/Helm for dev and staging.
OI-57	**Lab environments.** Academy / LMS	**Production needs: wildcard DNS and TLS, an egress firewall for labs, a longer random host part.** Live (24 Oct) on this laptop: the Academy signs each lab-service request for its own user (`X-Academy-User`, `X-Academy-Ts`, `X-Academy-Sig` = HMAC-SHA256 of user, time, method and path with `MISSION_SERVICE_SECRET`); mission-api runs with `MISSION_INTERNAL_ONLY=1`, so its old email-token sign-in answers 404. All 9 lab images are built; mission-api's 205 tests pass (incl. 9 real-container tests and the link test). The 6 live labs were played end to end in the browser. mission-api changes (backed up in `_backup-2026-10-22/mission-api`): the signed identity (main.py) and `LAB_PROXY_RELOAD` (docker_lab.py; Windows development needs proxy restarts). Production still needs: wildcard DNS and TLS for `*.ctf.yojo.com`, an egress firewall for labs, and a longer random host part (see mission-api README).

Decided, done or recorded for information 38

Decided by the owner, built, or an assumption documented in code. No action unless Yojo wants a change. The owner's 24 Oct decisions come first.

ID	Item	Details
OI-64	**Sign-up details locked; onboarding once; resume.** Academy / LMS	**Decided (owner, 29 Oct).** A student fills the sign-up details once (the email is verified first, so an existing account signs straight in). Filled fields are then locked: changing them answers `PROFILE_LOCKED`; empty fields can still be completed, and display name and leaderboard visibility stay free. Support or a Platform Admin changes locked details. A new (self-signed-up) student takes the career quiz or skips it, picks a team (Red/Blue/Green), and only then the Student Academy opens; this happens once. Existing students go straight to the portal. "My Resume" (all plans) and Placement Mission show a resume built from those details plus the quiz result and chosen team; only the headline is editable. Portals open in a new browser tab after sign-in.
OI-59	**Trainer portal completed (owner, 26 Oct).** Academy / LMS	Built: student 360 with tabs, remarks and private notes (never deleted, edits keep history), escalation and "ask admin to contact guardian" requests (admin resolves in Student Reports, trainer is notified), Lab Monitor with step progress, stuck alerts and nudges, quiz and question editor, deadline extensions and reports, trainer-started messages, mentoring outcomes, and My Feedback. Matrix: `docs/TRAINER-INTEGRATION.md`.
OI-60	**Trainer feedback rules.** Academy / LMS	**Decided (owner, 26 Oct): no feedback rules.** No time limit: a student may rate any class they attended (present or late) or mentoring session they had, once it has ended. Each student has one rating per session, which they can change (so averages cannot be stuffed); trainers see ratings without names and cannot edit them; admins see names for moderation.
OI-61	**Lab Monitor "stuck" rule.** Academy / LMS	**Decided (owner, 26 Oct): final.** A student is shown as stuck on a lab when it is open with no progress for 3 days, or after 5 or more tries on a step that is not done yet. Informational only; the trainer can send a nudge (one in-app notification per student, lab and day).
OI-62	**Public tools section.** Academy / LMS	**Approved (owner, 26 Oct).** Each of the 34 tools in "Hands-on with the tools security teams use" shows a short card on hover, keyboard focus or tap: what it is and which Yojo track uses it.
OI-11	**Which missing profile fields block other routes with 409 `PROFILE_INCOMPLETE`.** Platform	**Decided (owner, 24 Oct): keep as is.** `GET /me/missing-fields` lists them; nothing blocks.
OI-12	**Breached-password check (AUTH-R04: k-anonymity service or bundled list).** Platform	**Decided (owner, 24 Oct): keep the bundled list** plus the policy rules. Nobody is ever given a random password: staff choose their own when accepting an invite or resetting, and CTF players choose theirs at registration. Added: on the staff invite and reset pages an optional "suggest a strong one" button fills a random 16-character password (browser secure random) that the person can keep, change or ignore; the server checks it with the same policy and list.

ID	Item	Details
OI-14	**CTF participants and the LMS.** Platform	**Built (owner, 24 Oct).** A CTF participant joins the Academy on the same account: on the student sign-in ("No" path) they enter the CTF email, prove it with the emailed code, and add the student details (name, institution, consents). The account becomes a student account; the CTF username, password, authenticator and registrations stay. The CTF dashboard links to this (`login.html#join-academy`). Sessions stay separate: an LMS session cannot open `/ctf/*` and the CTF session keeps working. Audited as `auth.student.joined_from_ctf` .
OI-16	**Staff email domains** Platform	**Decided (owner, 24 Oct):** `STAFF_EMAIL_DOMAINS=yojosolutions.com` .
OI-25	**CTF login OTP tries.** Auth / CTF / Admin	**Decided (owner, 24 Oct): 3 tries** (shared OTP setting, 5 min).
OI-30	**Event dates.** Public website	**Done (owner, 24 Oct): the year is shown** ("18 Jan 2027", "08 Feb 2027") on the website, both in the built-in text and when dates come from the API. Change dates in `PATCH /admin/events/:id` .
OI-31	**Fake counters and claims on index.html: hero "Threats blocked / Learners in range" (random numbers), 100% placement assurance, 200+ hiring partners, internship seat counts, CTF countdown, "142 teams (sample)".** Public website	**Decided (owner, 24 Oct): unchanged until real data arrives.**
OI-32	**Job Hunter. There is no job-matching service.** Public website	**Decided (owner, 24 Oct): stays a demo.** Now labelled "Demo" with sample companies; the false text "deleted after 24 hours" is replaced by "your file stays in your browser and is never uploaded or stored", and "12 more roles, sign up to unlock" by "These are sample results". A real service (job sources, matching, upload storage with malware scanning) is a future decision.
OI-37	**Newsletter.** Public website	**Removed (owner, 24 Oct: not planned).** Endpoint, service code, consent version and test removed; migration `20261024140000_remove_newsletter` deletes any newsletter rows and drops the `newsletter` kind. Event reminders keep the table. Ask if a newsletter is wanted later.
OI-42	**Performance index and flags (trainer/admin batch overview).** Academy / LMS	**Decided (owner, 24 Oct): attendance 20 %, quiz average 40 %, labs cleared 40 %** (setting `policy.piWeights` ; migration `20261026090000_performance_index_weights` updated the stored setting). Parts without data are left out and the other weights scaled up. Flags unchanged: attendance under `policy.attendanceMinPct` (75), no activity for 7 days, quiz average under 40 %.

ID	Item	Details
OI-48	**Trainer course editing.** Academy / LMS	**Built (owner, 24 Oct).** Trainers edit and delete modules, lessons and FAQs of their batches' courses, and every edit or delete needs a reason from a list (Content correction, Outdated content, Duplicate, Restructuring the course, Other) plus a comment box (required for Other). Stored in the audit log (`trainer.course.*`); admins see a "Trainer changes" list with who, when and why on the course page (`GET /admin/courses/:id/changes`). **Updated (owner, 29 Oct): course builder and approvals.** Trainers create courses (they own them), add co-editors, assign their own batches (a batch runs one course; moving a batch that already runs one stays with a Platform Admin) and publish their own courses. Drafts and unpublished courses are edited directly. Once a course is published, every trainer edit or delete (details, modules, lessons, FAQs, unpublishing) becomes a change request (CCR-…) with the reason and comment: the live course stays unchanged until a Platform Admin approves it in Course Management → Change requests (the change is applied and the version goes up) or rejects it with a note the trainer is shown. Both decisions are audited and the trainer gets a bell message. Price and free/paid stay admin-only. Cover images wait for file storage (OI-45).
OI-49	**Placement weights.** Academy / LMS	**Decided (owner, 24 Oct):** resume, mock interview, internship and portfolio at 25 % each. The mock-interview part now combines interviews and **communication skill** equally: half = interviews done ÷ target (3), half = the average communication score the trainer gives (1–5) as %. The student and admin views show the communication score. Portfolio target 5.
OI-51	**Leaderboard divisions.** Academy / LMS	**Decided (owner, 24 Oct):** Bronze → Silver → Gold → Platinum → Diamond, everyone starts in Bronze. Weekly (IST), top 7 move up; bottom 5 move down only when more than 12 in the division earned XP; no XP = stay. The weekly board shows the division ladder, the student's zone (promotion, safe, demotion, not started) and a "How the league works" explanation.
OI-53	**Career-path rules.** Academy / LMS	**Decided (owner, 24 Oct): confirmed as built.** Optional; self-signed-up students are asked first, Yojo-origin students are not forced. Percentages = answers per track ÷ 15 × 100 rounded; tie order Defensive, Offensive, GRC. One take per 30 days; the path can be changed any time (audited).
OI-56	**Lab steps and hints (24 Oct).** Academy / LMS	The 9 labs are server missions with steps (`prisma/lab-content.ts`): INT-01 and REC-01 lesson + quiz, DOM-01 briefing + date, the six practical labs briefing + answer. Answers and quiz keys are stored only as salted hashes; the seed file holds the plain values (like mission-api did), so keep the repository private. **Lab steps and the 21 hints (3 per practical lab) approved by the owner on 24 Oct (final).**
OI-06	**Where the backend lives. The task named `C:\Users\reenaz\Downloads\Yojo academy\Yojo academy` ; that Windows path is not reachable from this session.** Platform	**Update: the backend now lives in the Windows folder on this laptop (yojo-backend-auth-ctf-lms/backend).** Delivered to the connected folder `/Users/mac/Downloads/Yojo academy/backend` (new folder; nothing existing was changed). Move it if the Windows copy is the source of truth.

ID	Item	Details
OI-07	**Existing `mission-api` (Python FastAPI + SQLite, own tokens). It overlaps with the NestJS architecture in the spec (missions, labs, sessions).** Platform	**Update: kept as an internal lab service behind the Academy sign-in (see OI-57).** Left untouched and running separately. Decide in M2/M6: port to NestJS, keep as an internal lab service behind the new auth, or retire.
OI-15	**Front-end link format for emailed tokens. The prototype pages do not call any API yet.** Platform	**Update: the front-end pages are now wired to the API (see OI-28).** Links use `PUBLIC_APP_URL/login.html#invite=…`, `#reset=…`, `#mfa-reenrol=…` (hash fragment, so tokens are not sent to servers or logs). Front-end integration is a later step.
OI-17	**Account lockout representation** Platform	Failed-login lock uses `locked_until` (timed) and keeps `status`; reactivating via `PATCH /admin/staff/:id/status` clears it. The `locked` status value is kept for a manual lock if Yojo wants one.
OI-18	**Staff role change endpoint** Platform	With a single admin role it now only changes a trainer's capability (trainer / trainer_mentor), with reason and fresh MFA.
OI-20	**Single admin role (user's instruction).** Auth / CTF / Admin	Only `platform` is accepted; DB CHECK on `users` and `staff_invites`; `role_permissions` holds one row per permission. The 4 enum values stay in the type so history rows remain valid. MFA reset still needs a second Platform Admin.
OI-22	**CTF events.** Auth / CTF / Admin	**Update: event admin (PATCH /admin/events/:id) was added in the public website stage (see OI-30).** One default event `yojo-ctf` (status `registration`) is seeded so registration works; event management APIs are not built.
OI-23	**CTF linking rule.** Auth / CTF / Admin	An existing student registering for the CTF with the same email is linked (no second user) only after the email OTP succeeds; audited as `ctf.identity.linked`. Staff emails are refused (no code is mailed). Username is global and case-insensitive.
OI-24	**CTF session.** Auth / CTF / Admin	`aud=ctf`, 8 h absolute, student idle timeout; may reach only `/ctf/*`, logout and session list. LMS sessions cannot open `/ctf/*`. Post-login redirect allow-list in `src/ctf/ctf-auth.service.ts`.
OI-26	**Admin LMS "pending onboarding".** Auth / CTF / Admin	Means staff who accepted an invite but have not finished MFA/policy/profile. Student onboarding completeness is exposed per student through `/me/missing-fields` only.
OI-27	**Admin LMS paging.** Auth / CTF / Admin	Opaque offset cursor (`nextCursor`), max 100 per page; fine for the expected size, switch to keyset if lists reach 100k+.

ID	Item	Details
OI-28	**Frontend wiring.** Auth / CTF / Admin	Stage 3 (22 Oct): `login.html`, the three CTF pages and the admin user lists are wired. 24 Oct: `academy.html` (all sections), and the admin Attendance, Performance, Notifications, Courses, Labs, Assessments, Analytics, Insights, Reports and Certificates pages. **26 Oct: all 11 trainer sections are live** (see `docs/TRAINER-INTEGRATION.md`). Four admin screens stay on demo data by the owner's choice (OI-58).
OI-29	**Mission/lab platform `yojo-ctf` (FastAPI, own PBKDF2 users).** Auth / CTF / Admin	**Update: joined to the Academy identity through the signed service link (see OI-57).** Untouched. It still has its own accounts; joining it to this identity (token exchange or shared session) is a separate decision.
OI-33	**Testimonials, mentor figures and blog posts in index.html are hard-coded.** Public website	Not copied to the database. An admin publishes verified items through `/admin/content`; testimonials need `consentConfirmed`, posts need `mentorApproved`, and mentor figures show only with `figuresVerified`. Until something is published the page keeps its built-in text.
OI-40	**Spam protection.** Public website	Hidden honeypot field, per-IP (10/h) and per-email/phone (5/h) limits, CSRF pre-session token, 256 kB body limit. No CAPTCHA: add one only if spam gets through.
OI-43	**Mission service link. The 9 missions (INT-01 … IMG-01) run in `mission-api` (FastAPI, own users, bearer tokens issued for any email without checking it).** Academy / LMS	**Update: done. The signed server-to-server link is live and XP is paid from the ledger (see OI-57).** Not changed. The Academy shows them with "Mission service offline". Needed: a server-to-server contract (start a lab for a user the Academy vouches for, report verified completion) so XP is paid from the ledger. Mission categories in the seed are a default to confirm.
OI-50	**Staff notification emails.** Academy / LMS	Only important notifications are emailed, and staff currently receive only message notifications (not important), so no staff email is sent. If staff emails are added, their link must open `trainer.html`, not the Academy.
OI-54	**Admin Courses, Labs and Assessments (owner, 24 Oct).** Academy / LMS	In live mode the admin console manages them again (they were trainer-only in the demo). Trainers keep content editing for their batches' courses (OI-48).

Known dependency advisories

- 2 moderate: `js-yaml` through `@nestjs/swagger` 11.x. Swagger is off in production (config refuses it). Fixed in `@nestjs/swagger` 12 (major upgrade, not done in M1).
- `deepmerge-ts` (high, in the Prisma CLI config loader) is pinned to the fixed 8.0.2 with an npm `overrides` entry; `prisma validate`, `generate` and `migrate` were re-run after the override and work. Remove the override when moving to Prisma 7.

Not in Milestone 1 (by instruction)

Courses, batches, attendance, labs and the range controller, CTF, AI, payments, notifications beyond the mail port, files, reports, DPDP export and deletion, metrics endpoint and dashboards.